Arash Sarshar

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EDUCATION

Ph.D. Computer Science, Virginia Tech, 2020

M.S. Electrical Engineering, Ferdowsi University, 2012

B.S. Electrical Engineering, Shahid Beheshti University, 2008

APPOINTMENTS

2023–present California State University, Long Beach
Assistant Professor, Department of Computer Engineering and Computer Science

2020–2022 Virginia Tech
Postdoctoral Associate, Department of Computer Science

RESEARCH AREAS

Scientific Machine Learning: Data-Driven and Equations-Based Modeling, Simulation, and Optimization

Computational Science: Numerical Methods for Forward and Inverse Problems, High-Performance Computing

PUBLICATIONS

Journal Articles

2026 A Pourkhalili, A Sarshar, L Margheim, C Rafie, "Association of Implementation Fidelity With Outcomes of a Distance Learning Lifestyle Change Program Delivered by Cooperative Extension," *Journal of the National Extension Association of Family and Consumer Sciences*, pp. 65–72. Article PDF.

2026 J Barry-Straume, AD Verulkar, A Sarshar, AA Popov, A Sandu, "Ensemble based Closed-Loop Optimal Control using Physics-Informed Neural Networks," *Communications in Nonlinear Science and Numerical Simulation*, 110063. https://doi.org/10.1016/j.cnsns.2026.110063

2025 A Raj, CE Gudumotou, S Bun, K Srinivasa, A Sarshar, "Deep Operator Networks for Bayesian Parameter Estimation in PDEs," *Computer Physics Communications*, 317, 109853. https://doi.org/10.1016/j.cpc.2025.109853

2025 J Barry-Straume, A Sarshar, AA Popov, A Sandu, "Physics-informed neural networks for PDE-constrained optimization and control," *Communications on Applied Mathematics and Computation*, 8, 1283–1306 (2026; online 2025). https://doi.org/10.1007/s42967-025-00499-x

2024 A. Bhattacharjee, A. Popov, A. Sarshar, A. Sandu, "Improving the Adaptive Moment Estimation (ADAM) stochastic optimizer through an Implicit-Explicit (IMEX) time-stepping approach," *Journal of Machine Learning for Modeling and Computing*, 2024. https://doi.org/10.1615/JMachLearnModelComput.2024053508

2022 S. Roberts, A. Popov, A. Sarshar, A. Sandu, "A fast time-stepping strategy for ODE systems equipped with a surrogate model," *SIAM Journal on Scientific Computing*, 44 (3), A1405-A1427. https://epubs.siam.org/doi/abs/10.1137/20M1386281

2022 S. González–Pinto, D. Hernández–Abreu, M. S. Pérez–Rodríguez, A. Sarshar, S. Roberts, A. Sandu, "A unified formulation of splitting-based implicit time integration schemes," *Journal of Computational Physics*, Volume 448, 2022. https://doi.org/10.1016/j.jcp.2021.110766

2021 S. Roberts, J. Loeffeld, A. Sarshar, C. S. Woodward, A. Sandu, "Implicit multirate GARK methods," *Journal of Scientific Computing*, Volume 87, Issue 1, 2021. https://doi.org/10.1007/s10915-020-01400-z

2021 A. Sarshar, S. Roberts, A. Sandu, "Alternating Directions Implicit Integration in a General Linear Method Framework," *Journal of Computational and Applied Mathematics*, Volume 389, 15 January 2021. https://doi.org/10.1016/j.cam.2019.112619

2020 S. Roberts, A. Sarshar, A. Sandu, "Coupled Multirate Infinitesimal GARK Schemes for Stiff Systems with Multiple Time Scales," *SIAM Journal on Scientific Computing*, Vol. 42, No. 3, PP A1609–A1638. https://epubs.siam.org/doi/abs/10.1137/19M1266952

2020 S. Roberts, A. Sarshar, A. Sandu, "Parallel Implicit-Explicit General Linear Methods," *Communications in Applied Mathematics and Computational Science*, Volume 6, Issue 2, 2021. https://doi.org/10.1007/s42967-020-00083-5

2019 A. Sarshar, S. Roberts, A. Sandu, "Design of high-order decoupled multirate GARK schemes," *SIAM Journal on Scientific Computing*, Vol. 41, No. 2, PP A816-A847. https://epubs.siam.org/doi/abs/10.1137/18M1182875

2017 A. Sarshar, P. Tranquilli, B. Pickering, A. McCall, A. Sandu, C. J. Roy, "A Numerical Investigation of Matrix-Free Implicit Time-Stepping Methods for Large CFD Simulations," *Computers & Fluids*, Volume 159, 15 December 2017, Pages 53–63. https://doi.org/10.1016/j.compfluid.2017.09.014

2017 P. Tranquilli, S. R. Glandon, A. Sarshar, A. Sandu, "Analytical Jacobian-vector products for the matrix-free time integration of partial differential equations," *Journal of Computational and Applied Mathematics*, Volume 310, 15 January 2017, Pages 213–223. http://www.sciencedirect.com/science/article/pii/S0377042716302199

Conference Proceedings

2016 A. Sarshar, A. E. Tabari, "Design of a Stacked Stub-Loaded Patch Element for X-band Reflectarray Antenna with True Time Delay," *7th European Conference on Antennas and Propagation*, Gothenberg, Sweden. http://ieeexplore.ieee.org/xpl/abstractAuthors.jsp?reload=true&tp=&arnumber=6546931

Manuscripts in Progress

2026 A Sarshar, "Multirate Stein Variational Gradient Descent for Efficient Bayesian Sampling." Under revision at *Journal of Computational Science*. arXiv preprint.

2026 A Sarshar, G Lucero, "PINN-Point: Adaptive Collocation Strategies for Physics-Informed Neural Networks on Complex 2D Geometries." Manuscript in preparation.

2026 L Vargas-Martinez, L B Thomas, A Sarshar, N Forouzesh, "Multi-Objective Optimization for Binding Free Energy Calculations." In preparation.

Technical Reports

2022 A. A. Popov, A. Sarshar, A. Chenault, A. Sandu, "A Meta-learning Formulation of the Autoencoder Problem for Non-linear Dimensionality Reduction," *Technical Report*, arXiv. https://arxiv.org/pdf/2207.06676

2019 A. Sarshar, S. Roberts, A. Sandu, "Linearly-Implicit General Linear Methods," *Technical Report*, Virginia Tech. https://vtechworks.lib.vt.edu/bitstream/handle/10919/109996/Technical-Report-Linearly-Implicit-GLM.pdf?sequence=1

INVITED TALKS

2025 "A Multirate Stein Gradient Variational Inference Method." California State University, Los Angeles.

2024 "Uncertainty Quantification for Scientific Machine Learning Using Deep Operator Networks." SIAM Annual Meeting 2024, Spokane, Washington. https://youtu.be/1iTHm9u9lro

2023 "Science-guided Machine Learning for Forward, Inverse, and Control Problems." Grand Valley State University, Michigan. https://people.cs.vt.edu/sarshar/talks/gvsu/

2023 "Machine Learning for Parameter Estimation of Partial Differential Equations." Engineering Horizons Seminars, College of Engineering, California State University, Long Beach.

2022 "On Time-stepping Methods for Gradient-flow Optimization." Canadian Applied and Industrial Mathematics Society, SCMAI 2022, Ontario, Canada. https://sarshar.dev/files/adam-gark-slides.pdf

2020 "Linearly Implicit Genral Linear methods." SIAM Conference on Computational Science and Engineering 2021, Atlanta, Georgia.

2019 "Discrete Multirate GARK schems." SIAM Conference on Computational Science and Engineering 2019, Spokane, Washington.
https://www.pathlms.com/siam/courses/10878/sections/14361/video_presentations/127463

CONFERENCE ACTIVITY

Sessions Organized

2024 "New Methods in Probabilistic and Science-Guided Machine Learning." SIAM Annual Meeting 2024, Spokane, Washington.

2021 "Advances in Time-Stepping Methods for Multiphysics and Multiscale Problems - Parts I & II." SIAM Conference on Computational Science and Engineering, Atlanta, Georgia.

GRANTS AND AWARDS

2024 CSULB ORED Multidisciplinary Research Grant (joint), " A Machine Learning-Based Multi-Scale, Multi-Fidelity Framework for Advanced Prediction of Thermodynamic Properties in Dense Gases."

GRANT APPLICATIONS

2026 PI, "Efficient Generative AI Inference for Engineering Simulation." CSULB Mini-Grant and Summer Stipend (MGSS) 2027–28.

2026 Co-PI, U.S. Department of Energy Genesis Mission proposal (DE-FOA-0003612). Contributed to large-language-model and agentic-AI development.

COURSES TAUGHT

California State University, Long Beach

Artificial Intelligence, Pattern Recognition, Machine Learning, Advanced Artificial Intelligence, Applied Machine Learning, Computer Simulation and Modeling, Deep Learning

Virginia Tech

Numerical Methods

Bahar Institute of Higher Education

Antennas and Wave Propagation

SERVICE

Service to the University

2026–present Faculty Representative, CSULB College of Engineering Dean's Cabinet. Provide faculty input in discussions of summer course planning and student progression through required courses.

2025–2026 Chair, CSULB CECS Tenure-Track Search Committee: Computer Engineering, Cybersecurity, and Software Engineering.

2025 CSULB COE Engineering Faculty Council. Helped develop the charges, membership provisions, and responsibilities of the new Graduate Studies and Budget Advisory committees.

2025 CSULB CECS RTP Revision Committee

2024 CSULB CECS/STATS Data Science Degree Committee

2024–2026 CSULB COE Awards Committee. In the 2025–2026 cycle, evaluated and ranked candidates in four college faculty-award categories and three university faculty-award categories using independent blind scoring; contributed recommendations to the Dean and Academic Senate.

2024, 2025 CSULB CECS Graduate Curriculum Committee

2023 CSULB CECS Undergraduate Curriculum Committee

Thesis Advising

2026 Angel Mendez, MS CS. Mathematical foundations and efficiency of flow-matching generative models.

2026 Sakol Bun, MS CS. Machine-learning surrogate models for computational fluid dynamics. Recipient of the 2026 University Achievement Award for Outstanding Graduate Student in Research, Scholarly and Creative Activity.

2024 Keerthana Srinivasa Naidu, MS CS.

2024 Amogh Raj, MS CS.

2023 Carol Gudumotu, MS CS.

Thesis Advisory Committees

2026 Akash Chowdary Kodali, MS CS. Committee member.

2026 Tommy Long, MS CS. Committee member; chair: Dr. Shabnam Sodagari.

2026 Michael Mancini, MS CS. Committee member; chair: Dr. Shabnam Sodagari.

2024 Rahul Deo Vishwakarma, "Uncertainty-Aware Multimodal Deep Learning for Guaranteed Coverage in Hardware Trojan Detection." Committee member; chair: Dr. Amin Rezaei.

2024 Jonathan Bernabe Dabu, "Fostering Fairness: Knowledge Graph Integration in the ASTRAEA Methodology." Committee member; chair: Dr. Muhammad Abdul Basit Ur Rahim.

PEER REVIEW

Academic Journal Peer Review

SIAM Journal on Scientific Computing

International Journal of Computer Mathematics
Computing and Visualization in Science
Geo-scientific Model Development
International Journal for Numerical Methods in Engineering
Journal of Applied Numerical Mathematics
Journal of Computational and Applied Mathematics
Mathematics and Computers in Simulation
International Journal of Applied Mechanics
ICML Workshop on the Synergy of Scientific and Machine Learning Modeling

MEMBERSHIPS

Society for Industrial and Applied Mathematics
Institute of Electrical and Electronics Engineers
Association for Computing Machinery

PROFESSIONAL DEVELOPMENT

2025 Beach Mentor Program Certificate, California State University, Long Beach.
2018 Graduate Certificate in Future Professoriate, Virginia Tech.

Updated November 2026